

MAT 523: Functions of a Real Variable I

Homework IV

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(1.)

Theorem

(a) Every real sequence has a monotone subsequence. (b) Use this to provide another proof of the Bolzano-Weierstrass Theorem.

Proof

Suppose $E \neq \mathbb{R}$. Then E must be bounded above, bounded below, or both. Without loss of generality, suppose E is bounded above. Then, since E is nonempty, it must have a supremum $\ell = \sup E$. Now, it is either the case that $\ell \in E$ or $\ell \notin E$. Suppose, first, that $\ell \in E$. Since E is open, there must be an open set $(a, b) \subseteq E$ such that $\ell \in (a, b)$. But for ℓ to be in (a, b) , ℓ must be less than b . But $b \in E$, which contradicts the fact that ℓ is an upper bound for E . Next, suppose that $\ell \notin E$. Let (c, d) be an open set containing ℓ . If $E \cap (c, d) = \emptyset$, then all elements of E must be less than or equal to c . But then c is an upper bound for E that is less than ℓ . This is a contradiction, so $E \cap (c, d) \neq \emptyset$. Thus, ℓ is a closure point of E . But $\ell \notin E$, meaning E is not closed. This is a contradiction. Therefore, $E = \mathbb{R}$.

■

(3.)

Theorem

A compact set in $\mathbb{R}$ is closed and bounded.

Proof

Let $E \subseteq \mathbb{R}$ be a compact set. If E is empty, then by Proposition 12, E is closed. It can also be argued vacuously that E is bounded above and below. Now suppose E is non-empty. Let $\mathcal{C}$ be an open cover of E . Since E is compact, $\mathcal{C}$ has a finite subcover of E , call it F . Since F contains finitely many open sets, we can write them all down

$$(a_1, b_1), (a_2, b_2), \dots (a_n, b_n)$$

so that $a_1 < a_2 < \dots < a_n$. Since $E \subseteq (a_1, b_1) \cup (a_2, b_2) \dots \cup (a_n, b_n)$ (by definition of an open cover), it is clearly bounded below by a_1 . We can also write down the open sets in F

$$(c_1, d_1), (c_2, d_2) \dots (c_n, d_n)$$

so that $d_1 < d_2 < \dots < d_n$. (Obviously, the elements of this list are identical to the elements of the list above. We simply use different notation to allow for the possibility of a different ordering.) Again, since $E \subseteq (c_1, d_1) \cup (c_2, d_2) \dots \cup (c_n, d_n)$, it must be bounded above by d_n . Since E is bounded above and below, E is bounded.

Next we show that E is closed. If not, then there exists an $x \in \overline{E}$ such that $x \notin E$. Since x is a closure point of E , there exists an (a, b) such that $x \in (a, b)$ and $E \cap (a, b) \neq \emptyset$. Thus, there is a $y \in E$ that is also in (a, b) . Let $\mathcal{C}$ be an open cover of E . Since E is compact, there is a finite subcover of E , call it $\mathcal{F}$.

(4.)

Problem

Show that the conclusions of the Nested Set Theorem can fail to hold if **(a)** instead of compact sets, we take sets that are merely closed. Likewise, show that it can fail if **(b)** the sets are taken to be open and bounded.

Solution

(a) For the first case, consider the sets

$$F_n = [n, \infty),$$

where $n \in \mathbb{N}$. Each F_n is closed, but unbounded (above). Furthermore, for all n , $F_{n+1} \subseteq F_n$. But the intersection

$$\bigcap_{n=1}^{\infty} F_n = [1, \infty) \cap [2, \infty) \cap [3, \infty) \cap \dots$$

is empty, by the following logic. Suppose there is some finite real number x that is in each of the F_n . Let $N \in \mathbb{N}$ be the smallest natural number bigger than x . Then $x \notin F_N = [N, \infty)$. But F_N is part of the countably infinite intersection above. So, in fact, $x \notin \bigcap_{n=1}^{\infty} F_n$, and $\bigcap_{n=1}^{\infty} F_n = \emptyset$.

(b) For the second case, consider the sets

$$F_n = \left(0, \frac{1}{n}\right),$$

where $n \in \mathbb{N}$. Each set is open and bounded (above by 1, and below by 0). Furthermore, for all n , $F_{n+1} \subseteq F_n$. But the intersection

$$\bigcap_{n=1}^{\infty} F_n = (0, 1) \cap (0, 1/2) \cap (0, 1/3) \cap \dots$$

is empty. Take any candidate x such that $x \in \bigcap_{n=1}^{\infty} F_n$. let $N \in \mathbb{N}$ be the largest natural number such that $\frac{1}{N} < x$. The set $(0, 1/N)$ is part of the countably infinite intersection above, but obviously $x \notin (0, 1/N)$, so $\bigcap_{n=1}^{\infty} F_n = \emptyset$.

(5.)

Theorem

Let E be a set of isolated points. Then E is countable.

Proof

If E is finite, then E is obviously countable, so suppose E is infinite. Since E consists only of isolated points, each $x \in E$ has an associated radius $r_x > 0$ such that

$$(x - r_x, x + r_x) \cap E = \{x\}.$$

Note that we can make the r_x sufficiently small so that none of the sets $(x - r_x, x + r_x)$ overlap each other. By making the r_x smaller we do not run the risk of including other points of E (besides x) in the interval $(x - r_x, x + r_x)$. Now, with the r_x defined so that no overlapping occurs, for each $x \in E$ define q_x to be some rational number in $(x - r_x, x + r_x)$. The existence of each q_x is guaranteed by the fact that $\mathbb{Q}$ is dense in $\mathbb{R}$. Let S be the set of all the q_x . By Theorem 3 from Royden and Fitzpatrick, S is countable, since $S \subseteq \mathbb{Q}$, which is known to be countable. Let $f : E \rightarrow S$ be defined by $f(x) = q_x$. Clearly, f is onto, since each $q_x \in S$ is defined in terms of an $x \in E$. To show that f is one-to-one, suppose $f(x) = f(y)$, where $x, y \in E$. Then $q_x = q_y$. We know that, if $x \neq y$, then $(x - r_x, x + r_x)$ and $(y - r_y, y + r_y)$ do not overlap, meaning it is impossible for q_x and q_y to be equal (since they are in two different, non-overlapping intervals). Thus, it must be the case that $x = y$, so f is one-to-one. Since f is one-to-one and onto, it is a bijection. Therefore, E is countable. ■